The Determinant from Multilinear Algebra

A unified introduction for first-year students

1. What Are We Trying to Measure?

Take n vectors $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n \in \mathbb{R}^n$. Together they span a *parallelotope*: in $\mathbb{R}^2$ this is a parallelogram; in $\mathbb{R}^3$, a parallelepiped. We want a function $\omega(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n)$ that measures the **signed volume** of this shape. "Signed" means we track orientation as well as size: scaling one edge by λ should multiply the volume by λ, and swapping two edges should flip the sign.

If you have seen determinants before, you already know the destination. The determinant of a 2×2 matrix gives the signed area of the parallelogram spanned by its columns, and the determinant of a 3×3 matrix gives the signed volume of the parallelepiped. What is less clear is *why* the formula takes the form it does. The goal here is to show that the determinant formula is the *only* answer consistent with a few natural requirements on signed volume — it is not conjured from thin air, but forced on us.

For those thinking more abstractly: a linear map $T : V \to V$ stretches, compresses, or flips space. The determinant should capture the signed volume-scaling factor of T as a property intrinsic to T itself, independent of any choice of basis. Both perspectives lead to the same three axioms.

2. Multilinear Maps

Our first requirement is that the signed-volume function be linear in each argument separately. Scaling one edge of the parallelotope by λ should scale the volume by λ. Splitting one edge into a sum $\boldsymbol{u} + \boldsymbol{v}$ should split the parallelotope into two, with volumes adding.

Definition (multilinear (or k-linear)). Let V be a vector space over a field F. A map $M : V^k \to F$ is ***k*-linear** if it is linear in each of its k arguments separately, with all others held fixed. That is, for each slot i and for all scalars $\lambda \in F$ and vectors $\boldsymbol{u}, \boldsymbol{v} \in V$:

$$M(\boldsymbol{v}_1, \cdots, \lambda\boldsymbol{u} + \boldsymbol{v}, \cdots, \boldsymbol{v}_k) = \lambda, M(\boldsymbol{v}_1, \cdots, \boldsymbol{u}, \cdots, \boldsymbol{v}_k) + M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}, \cdots, \boldsymbol{v}_k),$$

where $\lambda\boldsymbol{u} + \boldsymbol{v}$ occupies the i-th position.

Example. The dot product $(\boldsymbol{u}, \boldsymbol{v}) \mapsto \boldsymbol{u} \cdot \boldsymbol{v}$ on $\mathbb{R}^n$ is bilinear (2-linear): linear in each factor independently. The familiar 2×2 determinant $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, viewed as a function of its two column vectors, is also bilinear: scale either column by λ and the output scales by λ.

Multilinear maps form a vector space: you can add two of them pointwise and scale them by constants, and the result remains multilinear. This will matter when we count how many signed-volume functions there can be.

3. The Alternating Condition

Multilinearity alone does not determine a unique signed-volume function. We also need to enforce orientation. Swapping two edges of a parallelotope should reverse its orientation, changing the sign of the signed volume. And if two edges coincide — a degenerate configuration — the volume should be zero.

> **Definition (alternating).** A k-linear map $M : V^k \to F$ is **alternating** if it vanishes whenever any two arguments are equal:
>
> $$M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_k) = 0 \quad \text{whenever } \boldsymbol{v}_i = \boldsymbol{v}_j \text{ for some } i \neq j.$$

This single condition implies antisymmetry. To see why: consider expanding $M\big(\cdots, \boldsymbol{v}_i + \boldsymbol{v}_j, \cdots, \boldsymbol{v}_i + \boldsymbol{v}_j, \cdots\big)$ using multilinearity, once distributing in the i-th slot and once in the j-th. Both expansions must equal zero (equal arguments), and comparing them forces

$$M\big(\cdots, \boldsymbol{v}_i, \cdots, \boldsymbol{v}_j, \cdots\big) = -M\big(\cdots, \boldsymbol{v}_j, \cdots, \boldsymbol{v}_i, \cdots\big).$$

Swapping any two arguments negates the output. More generally, permuting all n arguments by a permutation $\sigma \in S_n$ multiplies M by the *sign* of σ:

$$M\big(\boldsymbol{v}_{\sigma(1)}, \cdots, \boldsymbol{v}_{\sigma(n)}\big) = \operatorname{sgn}(\sigma) \cdot M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n),$$

where $\operatorname{sgn}(\sigma) = +1$ if σ can be written as a product of an even number of transpositions, and -1 if odd.

A further consequence: if $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n$ are **linearly dependent**, then $M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n) = 0$. Any dependence relation lets one vector be written as a linear combination of the others; multilinearity expands M into a sum of terms, each containing a repeated argument, and the alternating condition kills every one of them. Geometrically, a degenerate parallelotope — whose edges fail to span all of $\mathbb{R}^n$ — has zero volume.

> **Remark.** In matrix language: linearly dependent columns mean the matrix is singular. The alternating condition is precisely what forces singular matrices to have determinant zero. Note that this is a *consequence* of the axioms, not a separate assumption.

4. The Key Theorem

We are now ready for the heart of the argument.

> **Theorem (one-dimensionality of alternating n-linear forms).** Let V be an n-dimensional vector space over a field F. The space of all alternating n-linear maps $\omega : V^n \to F$ is **one-dimensional**: any two such maps are scalar multiples of each other.

Proof sketch. Fix a basis $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n$ of V, and let $M : V^n \to F$ be alternating and n-linear. Write each argument in basis coordinates: $\boldsymbol{v}_j = \sum_{i=1}^{n} a_{ij}\boldsymbol{e}_i$. Expanding $M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n)$ by multilinearity gives a sum over all n^n index tuples $(i_1, \cdots, i_n)$ of terms

$$a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \cdot M\left(\boldsymbol{e}_{i_1}, \cdots, \boldsymbol{e}_{i_n}\right).$$

The alternating condition eliminates most terms. Whenever two indices i_k coincide, two arguments are the same basis vector, so M vanishes. Only tuples where $(i_1, \cdots, i_n)$ is a permutation $\sigma \in S_n$ of $(1, \cdots, n)$ survive; there are exactly $n!$ of these. **Antisymmetry handles the rest.** For a surviving term,

$$M\left(\boldsymbol{e}_{\sigma(1)}, \cdots, \boldsymbol{e}_{\sigma(n)}\right) = \operatorname{sgn}(\sigma) \cdot M(\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n).$$

Substituting back:

$$M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n) = \left(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n} \right) \cdot M(\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n).$$

The sum in parentheses depends only on the coordinate entries a_{ij}, not on M at all. So M is completely determined by the single scalar $M(\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$: change that scalar, and M changes proportionally everywhere, but its shape is fixed. Any two alternating n-linear maps that agree on $(\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$ must agree everywhere.

This is what makes the determinant *inevitable*. There is, up to a scalar multiple, only one function consistent with multilinearity and the alternating property. Fixing that scalar — by specifying the volume of the unit hypercube — pins down the determinant uniquely.

5. The Determinant: Two Perspectives

The one-dimensionality theorem suggests two equivalent definitions, each illuminating a different facet.

5.1. Via normalization

Require that the unit hypercube — the parallelotope spanned by the standard basis vectors $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n$ — have signed volume 1. The formula from the proof sketch immediately becomes a complete and explicit definition.

Definition (determinant — Leibniz formula). The **determinant** of an $n \times n$ matrix $A = (a_{ij})$ is the unique alternating n-linear form in the columns of A satisfying $\det(I_n) = 1$. Its value is given by the **Leibniz formula**:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)},$$

where the sum runs over all $n!$ permutations σ of $\{1, \cdots, n\}$.

Example. For $n = 2$, there are two permutations: the identity ($\operatorname{sgn} = +1$) and the single transposition ($\operatorname{sgn} = -1$). The formula gives $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For $n = 3$, all six permutations produce the standard cofactor expansion familiar from a first linear algebra course.

This perspective is closest to computation, and it makes checking the formula against known cases immediate.

5.2. Via linear maps

For those thinking in terms of abstract linear maps $T : V \to V$: given any alternating n-linear form M, define the **pullback** of M along T by

$$(M \circ T)(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n) := M(T\boldsymbol{v}_1, \cdots, T\boldsymbol{v}_n).$$

Because T is linear and M is n-linear, $M \circ T$ is again n-linear. Because T is linear and M is alternating ($M = 0$ when two inputs coincide), $M \circ T$ is again alternating. So $M \circ T$ belongs to the same one-dimensional space as M. It must therefore be a scalar multiple of M: there exists $\lambda \in F$ with

$$M(T\boldsymbol{v}_1, \cdots, T\boldsymbol{v}_n) = \lambda \cdot M(\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n) \quad \text{for all } \boldsymbol{v}_1, \cdots, \boldsymbol{v}_n \in V.$$

If we replace M by any nonzero scalar multiple cM, both sides scale by c and λ is unchanged. So λ depends only on T, not on the choice of M.

Definition (determinant — intrinsic form). The **determinant** of a linear map $T : V \to V$ is the unique scalar $\det T \in F$ such that

$$M \circ T = (\det T) \cdot M$$

for every alternating n-linear map $M : V^n \to F$.

The two definitions agree: evaluating the intrinsic form at $(\boldsymbol{e}_1, \cdots, \boldsymbol{e}_n)$ with any nonzero M and applying the basis expansion recovers the Leibniz formula. The intrinsic form has the advantage of making certain properties transparent, as the next section shows.

6. Properties as Theorems

A key philosophical point of this approach: the familiar properties of the determinant are not additional axioms imposed by fiat. They are *theorems*, forced by the three axioms of multilinearity, the alternating property, and normalization. The table below records why each holds.

Property	Why it follows from the axioms
$\det(AB) = \det A \cdot \det B$	Both sides are alternating n-linear forms in the columns of A, normalized to agree on I_n; one-dimensionality forces equality. From the ab-

Property	Why it follows from the axioms
	stract side: $(M \circ S) \circ T = M \circ (ST)$, and tracking scalars gives $\det(S) \cdot \det(T)$.
$\det(A) = 0 \leftrightarrow A$ is singular	Singular $\Rightarrow$ linearly dependent columns $\Rightarrow$ the alternating form vanishes. Conversely, if A is invertible then $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$, so $\det(A) \neq 0$.
$\det(A^\top) = \det(A)$	Transposing swaps row and column indices. In the Leibniz sum, relabeling σ by σ^{-1} (same sign) shows the two sums are identical.
Scaling a row by λ multiplies det by λ	Scaling one row is a scaling of one argument — exactly what multilinearity governs.
Swapping two rows negates det	Direct consequence of antisymmetry. Adding a multiple of one row to another leaves det unchanged: the extra term has a repeated row, so it vanishes by the alternating property.

7. The Bigger Picture

The multilinear algebra framework does more than derive a formula — it locates the determinant within a much larger structure. The space of alternating n-linear forms on an n-dimensional space V is the **top exterior power** $\Lambda^n V^*$, and the Leibniz formula is the coordinate expression of the evaluation of this exterior form. The same ideas appear throughout mathematics:

- **Differential forms and integration.** The change-of-variables formula $\int_U f = \int_{\varphi(U)} (f \circ \varphi)\, |\det D\varphi|$ is the determinant acting as a volume-scaling factor on differential forms — the same signed-volume intuition, now on curved spaces.
- **Orientation on manifolds.** Choosing an orientation on a smooth manifold is precisely choosing a consistent nonzero element of $\Lambda^n T_p^* M$ at each point — the top exterior power of the cotangent space.
- **Algebra.** The same exterior-power construction applies to free modules over any commutative ring, making the determinant a concept in commutative algebra and algebraic geometry, not just in linear algebra over fields.

The entire structure flows from three axioms: **multilinearity, the alternating property, and normalization.** Everything else — the formula, multiplicativity, the relationship to invertibility and orientation — is forced. The determinant is not a definition; it is the unique answer those axioms demand.